

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
MID SEMESTER EXAMINATION, 2023
SESSION: 2025 – 2026 (Autumn)

Subject code: CS3009
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Subject Name: Operating Systems
Full Marks: 30

Dept. Code: CS
Duration: 2 Hrs

Note: Answer all Questions. Questions can be answered in any serial order. However, ensure that all parts of a question are answered sequentially.

1. (a) How is multiprogramming different from time-sharing systems? Do either of these require multiple processors? Justify your answer. [2+1][CO1, L1]
(b) Differentiate between Zombie and Daemon processes. Which of these starts at system boot and runs until shutdown? Which one is considered more harmful to the system, and why? [1+1+1] [CO2, L1]
(c) Draw and explain the Process State Diagram of an operating system. Describe the role of each state. Also, explain with an example how a process transitions from the Running state to the Waiting state. [1+2+1] [CO1, L1]
2. (a) In a multiprogramming environment, the convoy effect may occur during CPU scheduling. Explain what the convoy effect is, and analyze whether increasing the number of processors in the system will necessarily eliminate this problem. Justify your answer with reasoning. [2+2] [CO2, L2]
(b) Consider three processes A, B, C arriving in the ready queue at time 0 in the order A, B, C. Their service time requirements (including CPU and I/O) are 10, 20 and 30 units respectively. Hypothetically assume that each process spends 10% of its Service time on I/O followed by 70% of its Service time on Computation at CPU and last 10% on I/O before completion. Assuming Non-Concurrent I/O and Scheduling/Context switch Overhead = 1 unit. Draw the Gantt chart showing the execution schedule of these processes if the system follows FCFS (First Come First Serve) Scheduling policy. Also calculate the following: [2+2+2] [CO2, L3]
 - Average Turn-Around Time (TAT) of Processes
 - Average Waiting Time (WT) of Processes
 - CPU idleness
3. (a) Consider two processes, **P1** and **P2**, that both update a shared variable **X**. **P1** increments **X** by 1, while **P2** multiplies **X** by 2. Initially, **X** = 10. If proper synchronization is not used, what are the possible final values of **X** after both processes finish? Explain why this situation represents a Race Condition and define the Race Condition. [2+2] [CO2, L2]
(b) Consider the following algorithm that provides a solution to the two process critical section problem. Initially both the shared variables flag[0] and flag[1] are set to false. [2+2+2] [CO3, L3]

```
while(true){
    flag[0] = true;
    while(flag[1]){
        flag[0]= false;
        while(flag[1]);
        flag[0]= true;
    }
    {CRITICAL SECTION}
    flag[0]= false;
    {REMINDER SECTION}
}
```

Explain if the solution satisfies the properties: **Mutual Exclusion**, **Progress**, and **Bounded Wait**.